X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-09-01

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:
    - *CSP Partial cross section data*
    - *CST Temperature dependent cross section*
    - *ETA Average neutron yield per nonelastic event*
    - *NU Fission-neutron yield, nu-bar*
    - *TKE Total kinetic energy*
    - *SIG Cross section data given with limit as DATA and DATA-MIN*
    - *SIG Production cross section, where product coded as SF4 or in DATA (ELEM/MASS)*

    -- Currently implemented search: by full-Reaction-code, DatasetID, Author1
2. Plotting partial cross sections from local EXFOR and remote ENDF databases
    - *CSP vs ENDF MT:51/MF:3,33 - Production of a neutron, with residual in the 1st excited state*
    - *NU vs ENDF MT452,455,456/MF:1,31 - Average number of total/delayed/prompt neutrons released per fission event*
    - *ETA vs ENDF MT:452,18,102/MF:1,3,31,33 - Average neutron yield per nonelastic event: $\eta \pm \Delta\eta$*

Code reac1.py help:

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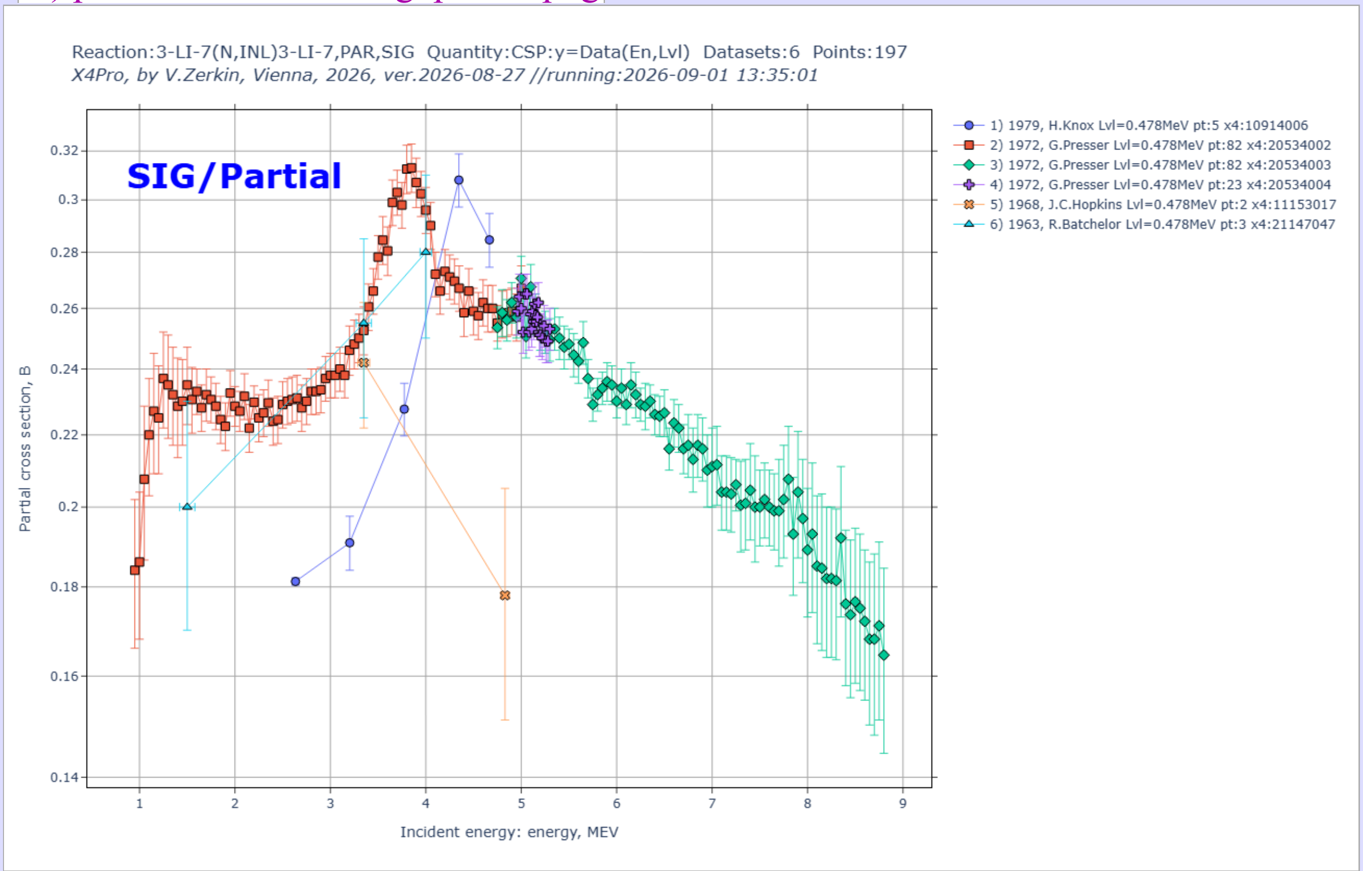
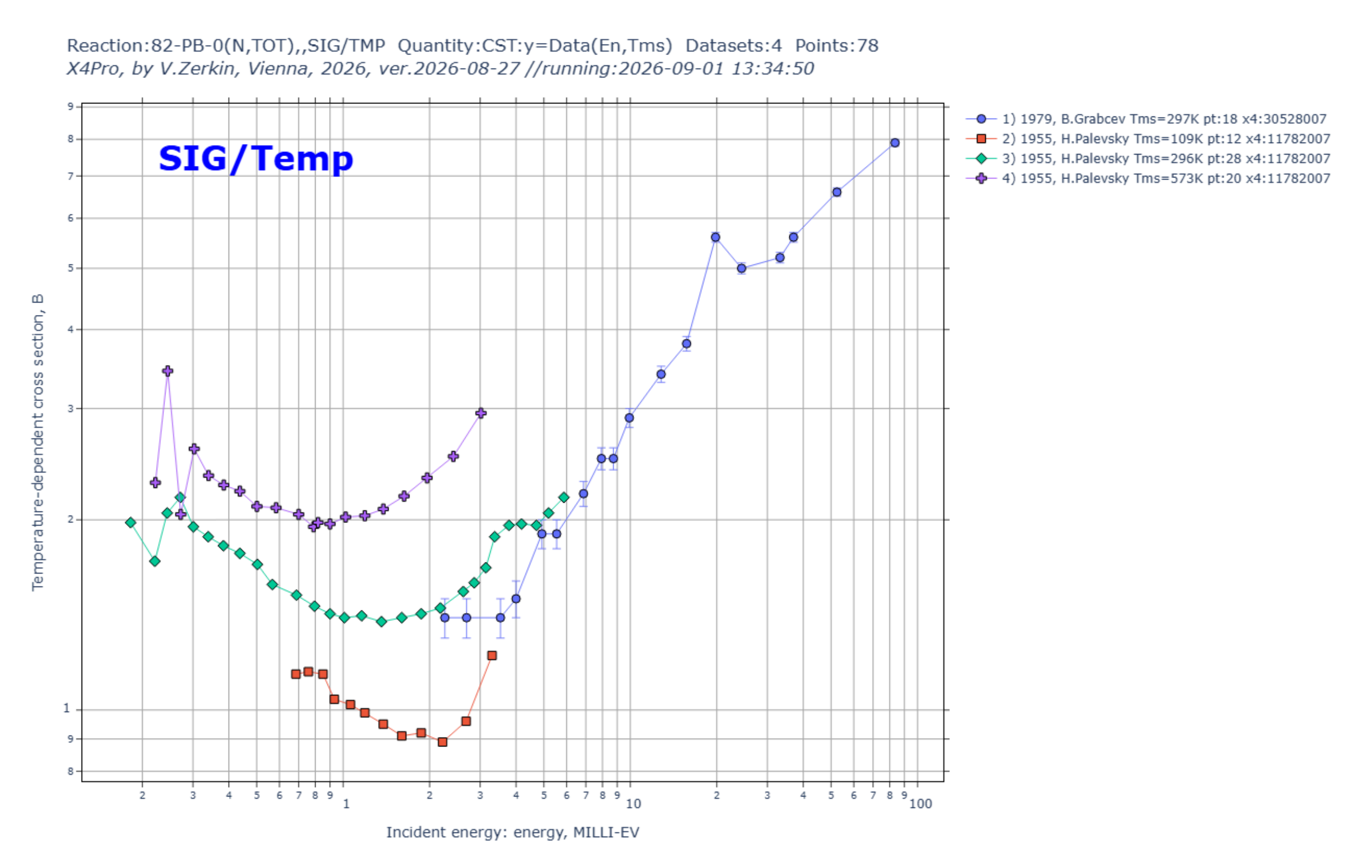
Program: reac1.py, ver.2026-08-14
Author:  V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
  * computational data are presented as function y=f(x1,x2,..)
    where f is measured quantity, xi are independent variables:
       EN  - incident energy,                 E2  - energy of outgoing particle,
       ANG - angle of outgoing particle,       EVL - energy level, etc.
Algorithm:
  * user defines EXFOR Reaction-code for data search, x-variable by number
  * selects x-variable by number (default: x1)
  * other variables will be used as parameter for grouping of data points
Usage:
    $ python [{flag}] x4update.py {[option|reaction]}
      * flag: see all Python flags:  $ python --help
        -B          don't write .pyc files on import
      * option:
        -help         print this help-text and exit (also --help)
        -o:<file>     output file name
        -x:x?         set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
        -x?fam:<str>  filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
        -x?min:<num>  filter out values of x? variable, e.g. -x1min:6e6
        -x?max:<num>  filter out values of x? variable, e.g. -x1max:30e6
        -nogrp        avoid groupping datasets by reaction-codes on the plot
        -fx:<num>     set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
        -fy:<num>     set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
        plotting:
        -xlog         logarithmic scale of X-Axis
        -ylog         logarithmic scale of Y-Axis
        -lines        connect points on the plot by lines
        -sym          draw border of a symbol of a data point on the plot
        -annot:x,y,txt  annotation on top of the plot;
                        x and y should be given in units of the plot;
                        text can include html tags: <b>, <sup>
      * reaction:
        <reaction>    full EXFOR Reaction-code, e.g. "8-O-16(N,EL)8-O-16,,DA"
```

part1-8-reac/ *EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA*

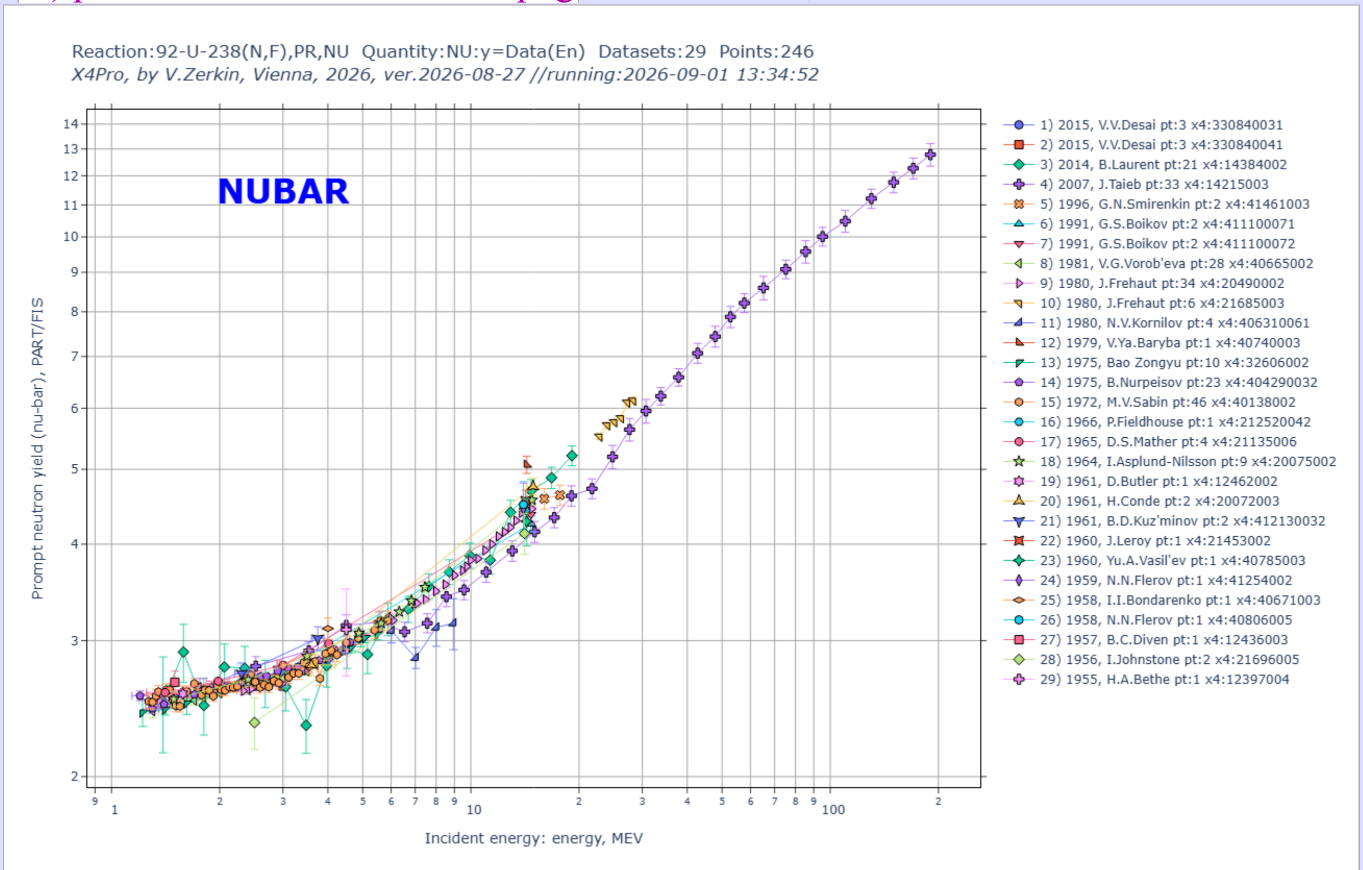
1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data



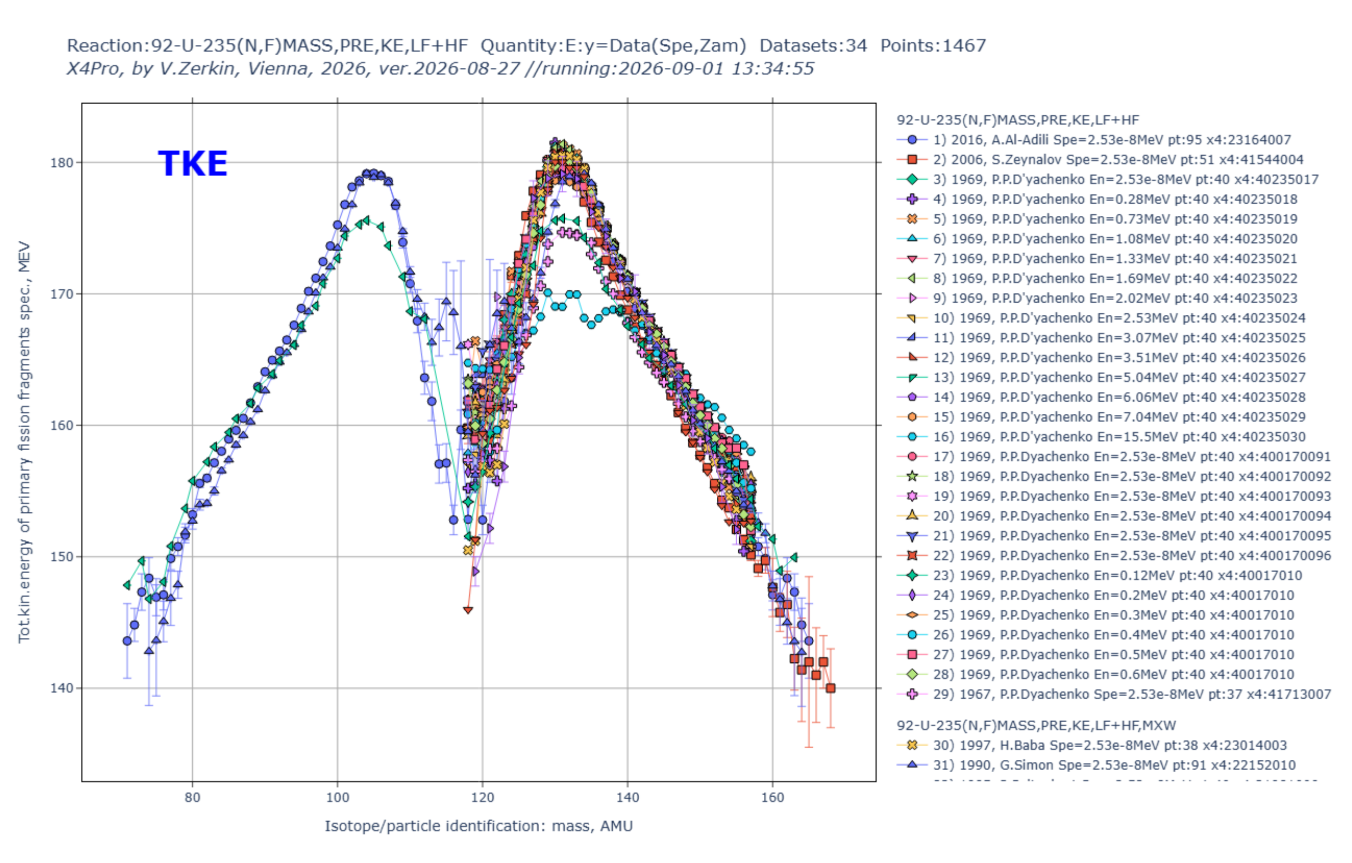
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section



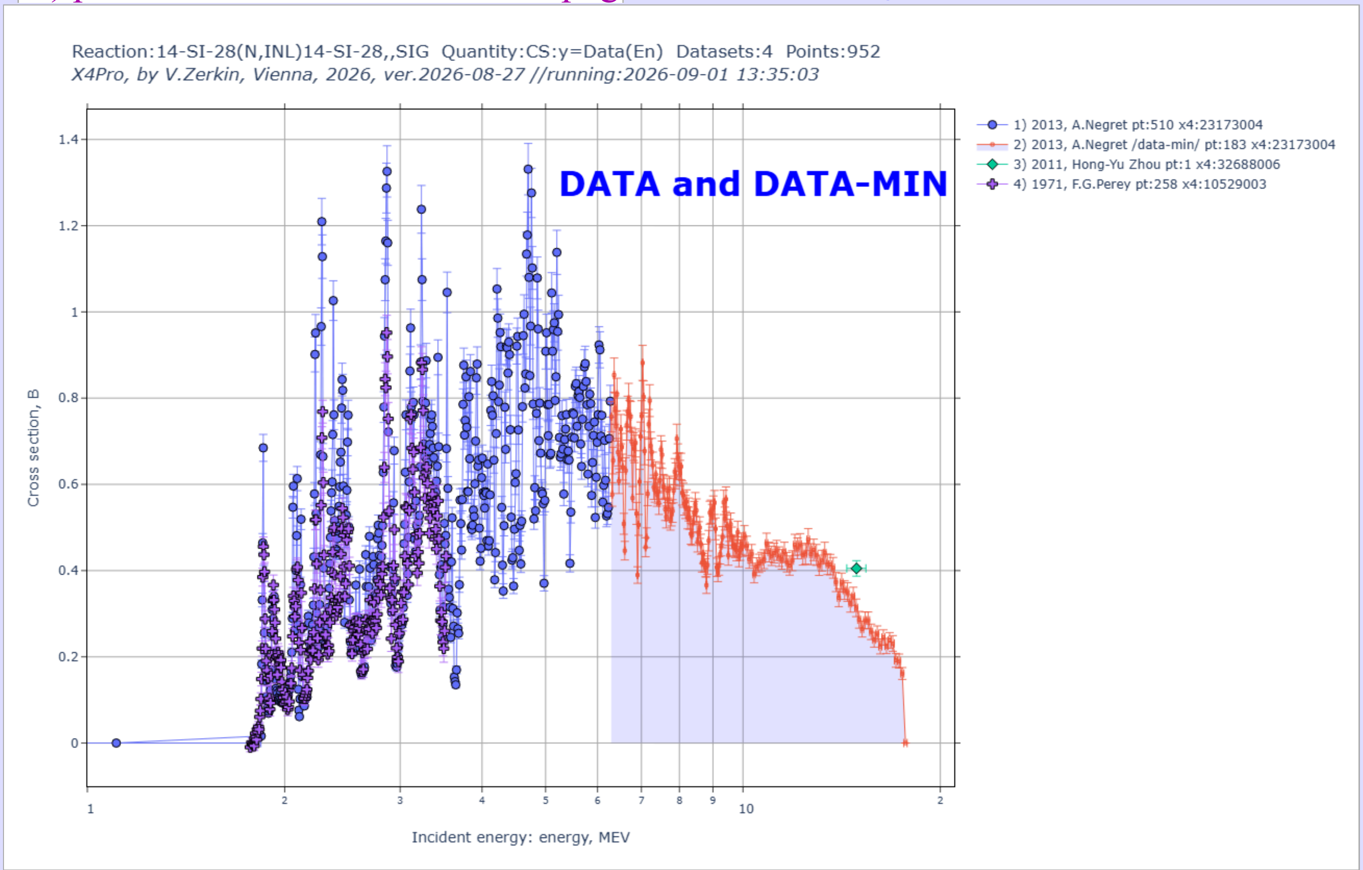
3) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar



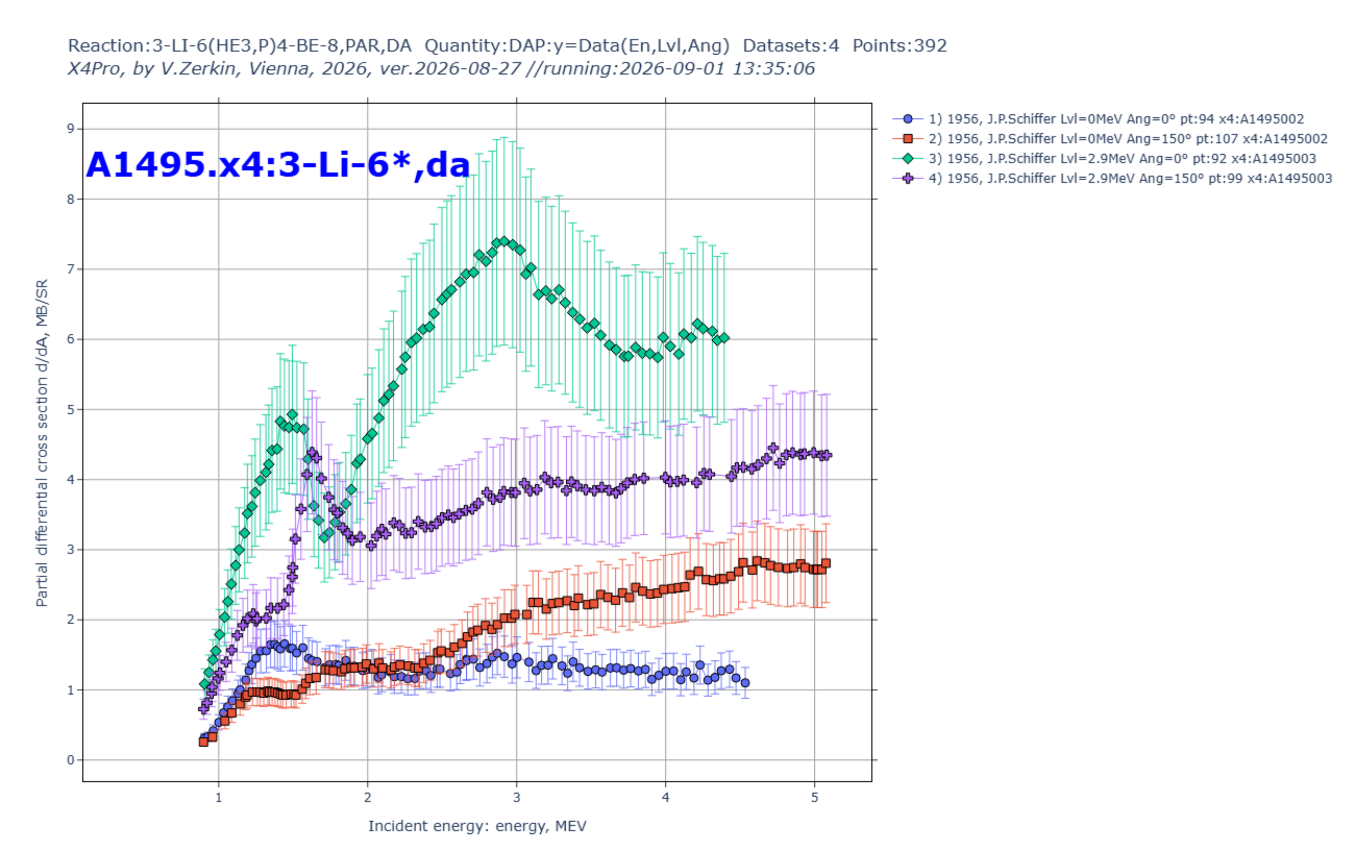
4) part1-8-reac/out00/tke1.html.png TKE Total kinetic energy



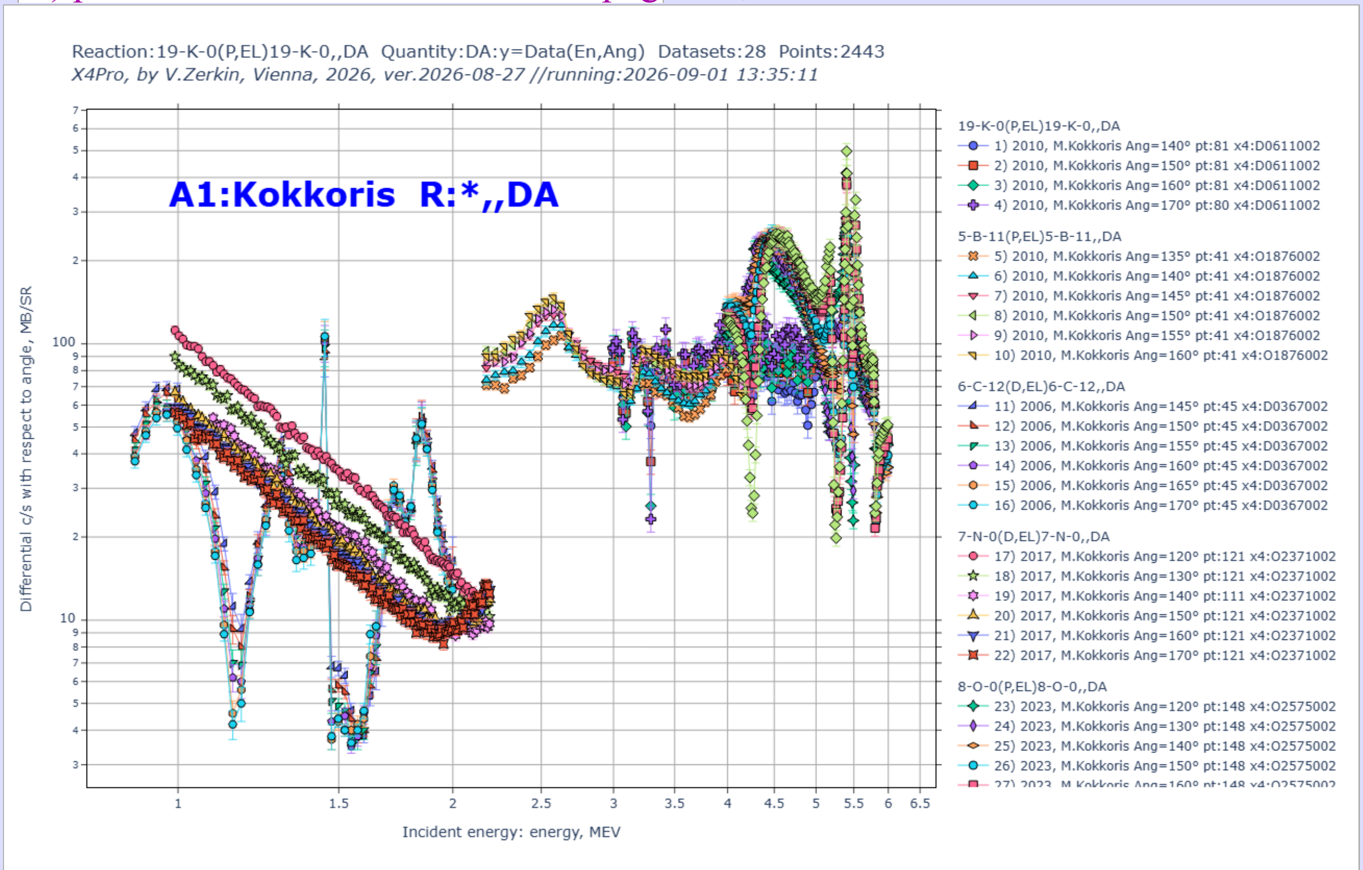
5) part1-8-reac/out00/si28nn.html.png SIG Cross section data given with limit as DATA and DATA-MIN



6) part1-8-reac/out00/a1495-li6.html.png Data from EXFOR:"A1495" & Reaction: "3-LI-6*DA"

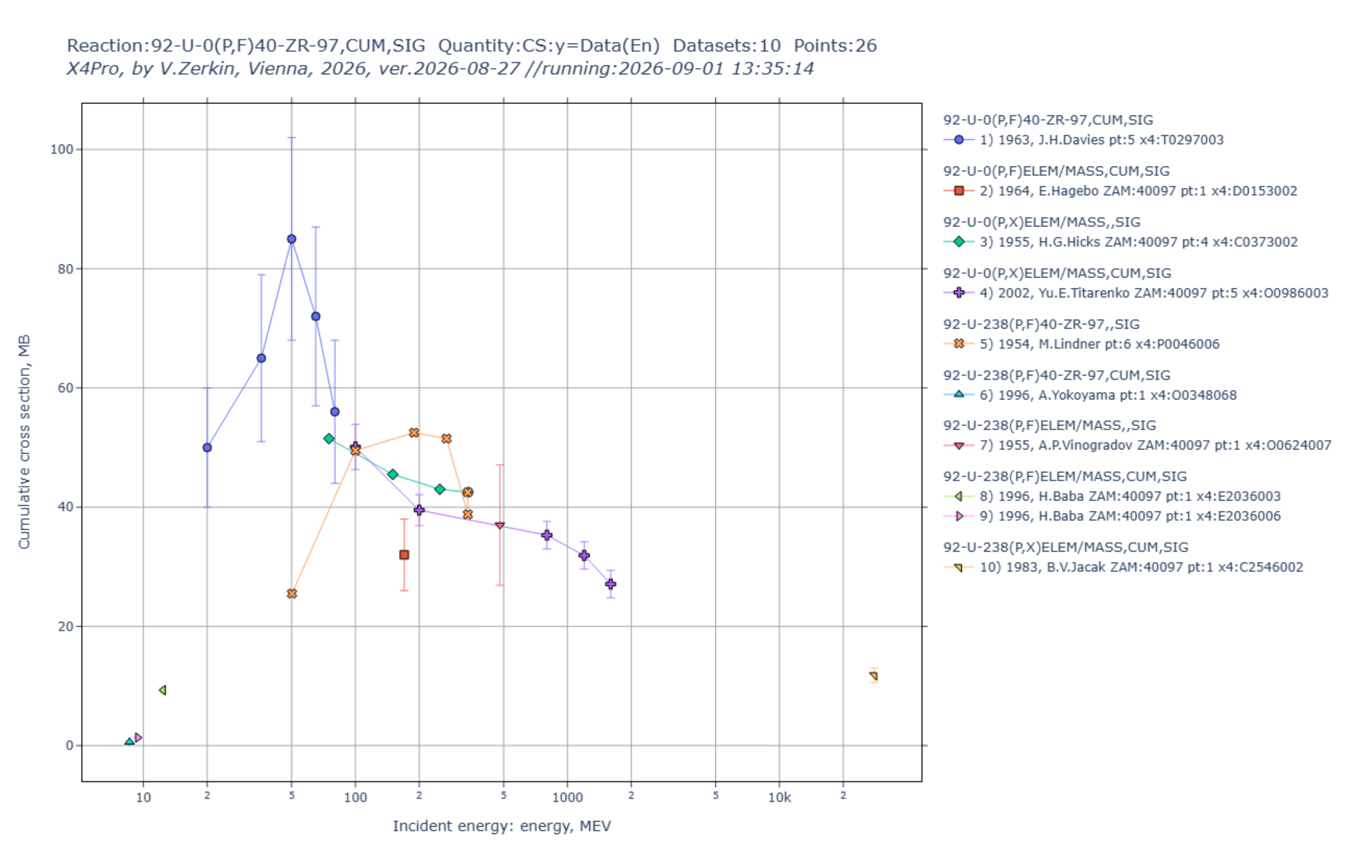


7) part1-8-reac/out00/Kokkoris.html.png Data by "Kokkoris" & Reaction: "*,,DA"



8) part1-8-reac/out00/sig1prod.html.png

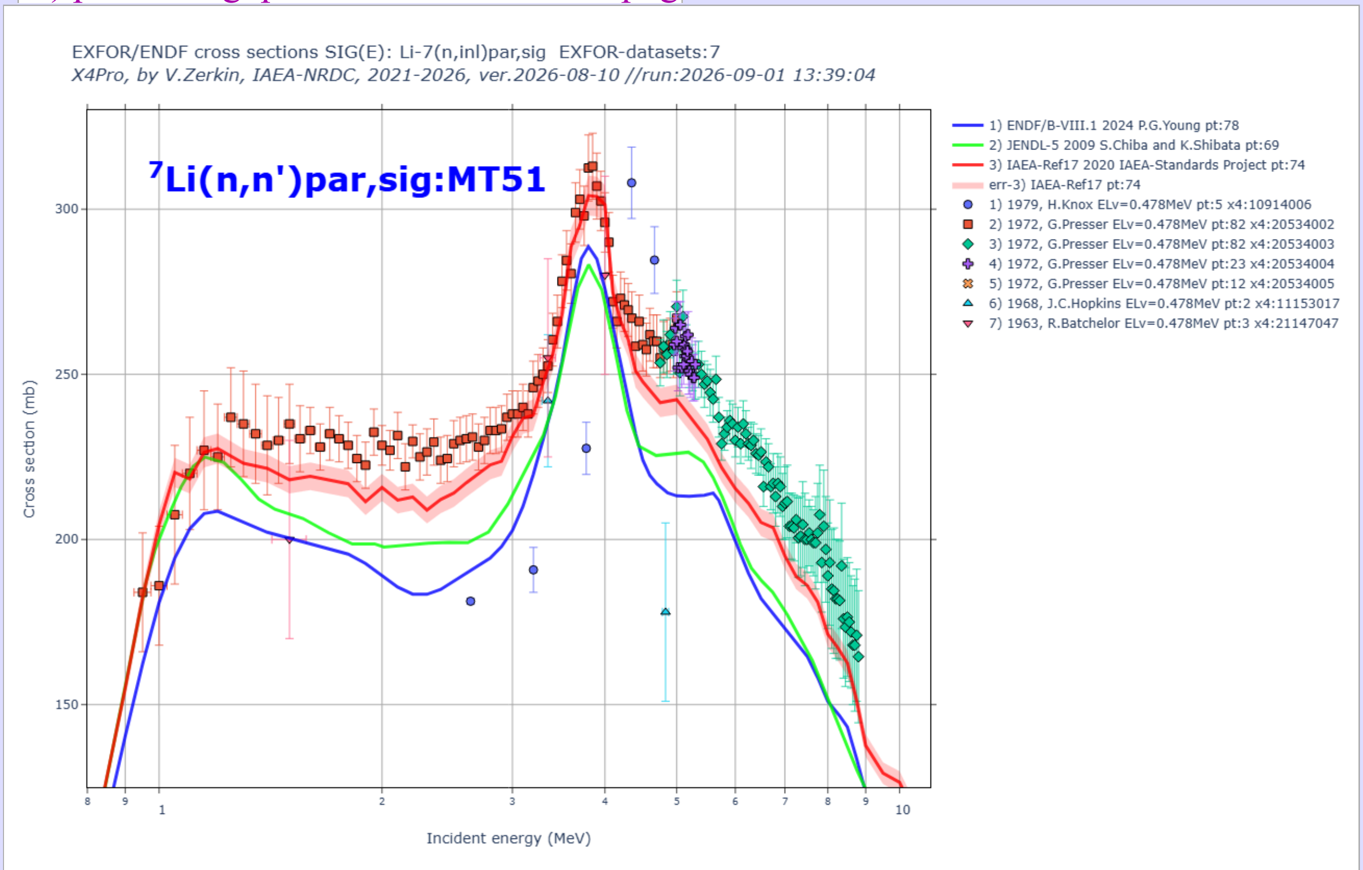
Production cross section coded as SF4 and ELEM/MASS



part2-8-sig1par/ *EXFOR/ENDF partial cross sections CSP:MT51*

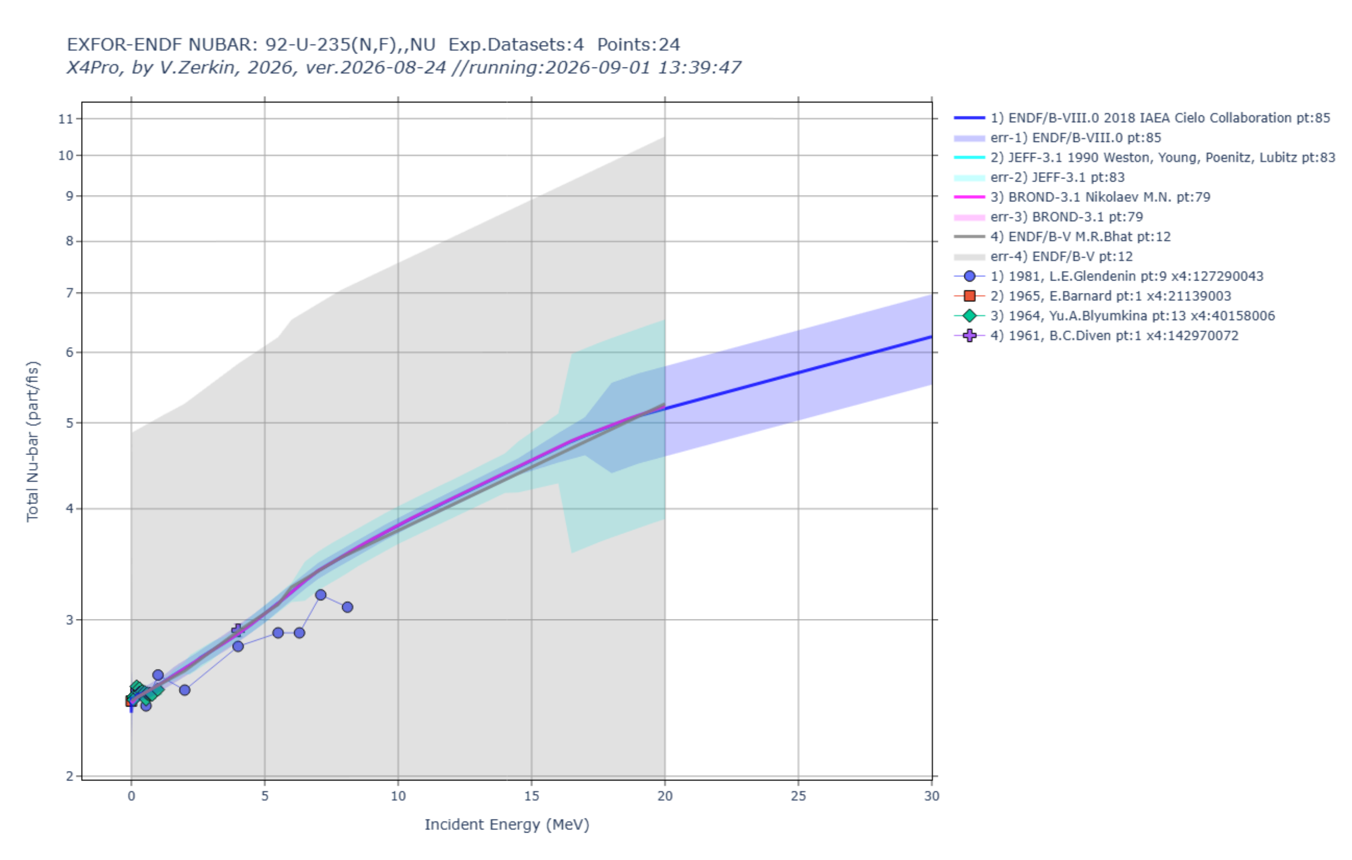
9) part2-8-sig1par/out00/Li7ninl.html.png

Production of a neutron, with residual in the 1st excited state

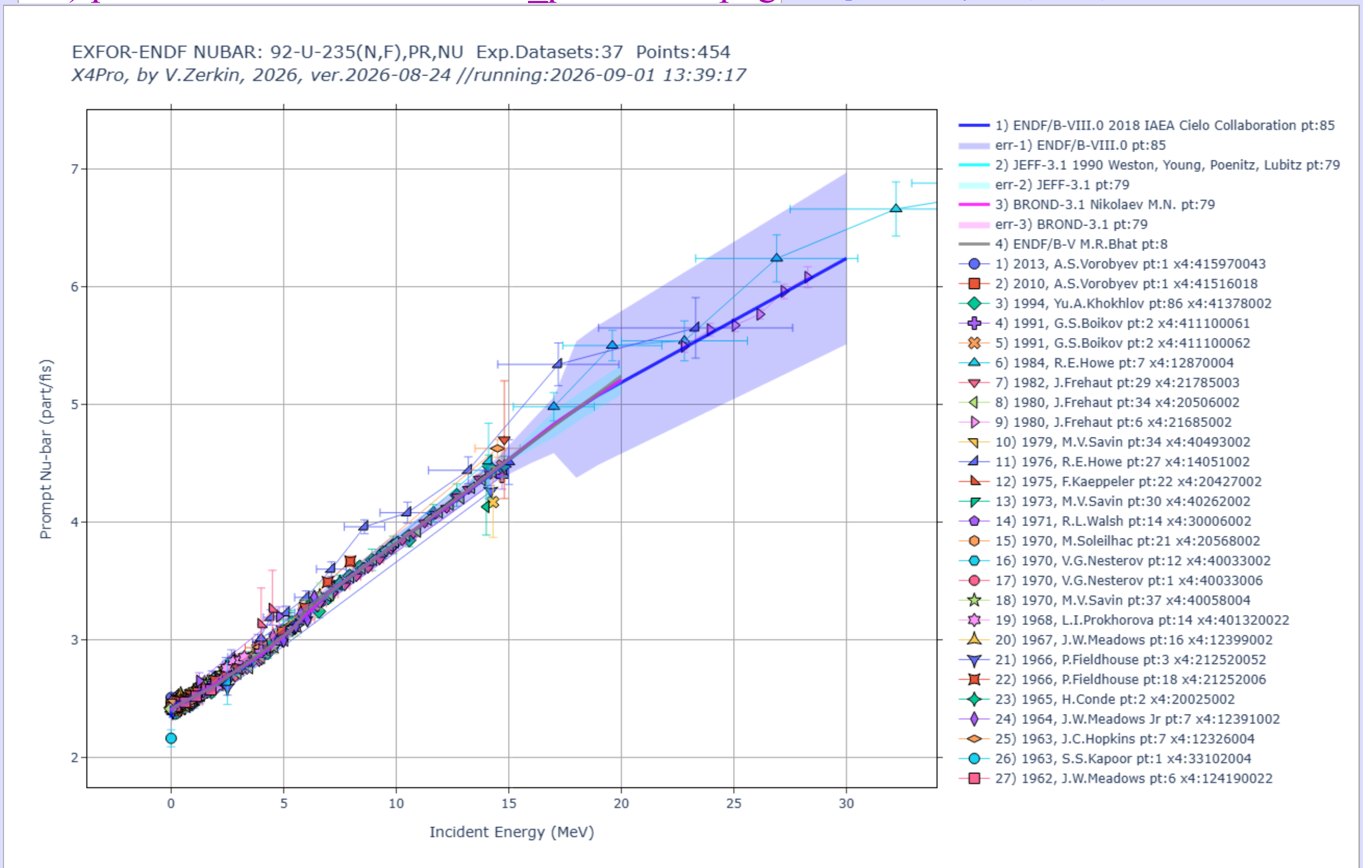


part2-9-nubar/ *EXFOR/ENDF NUBAR - average number of neutrons per fission*

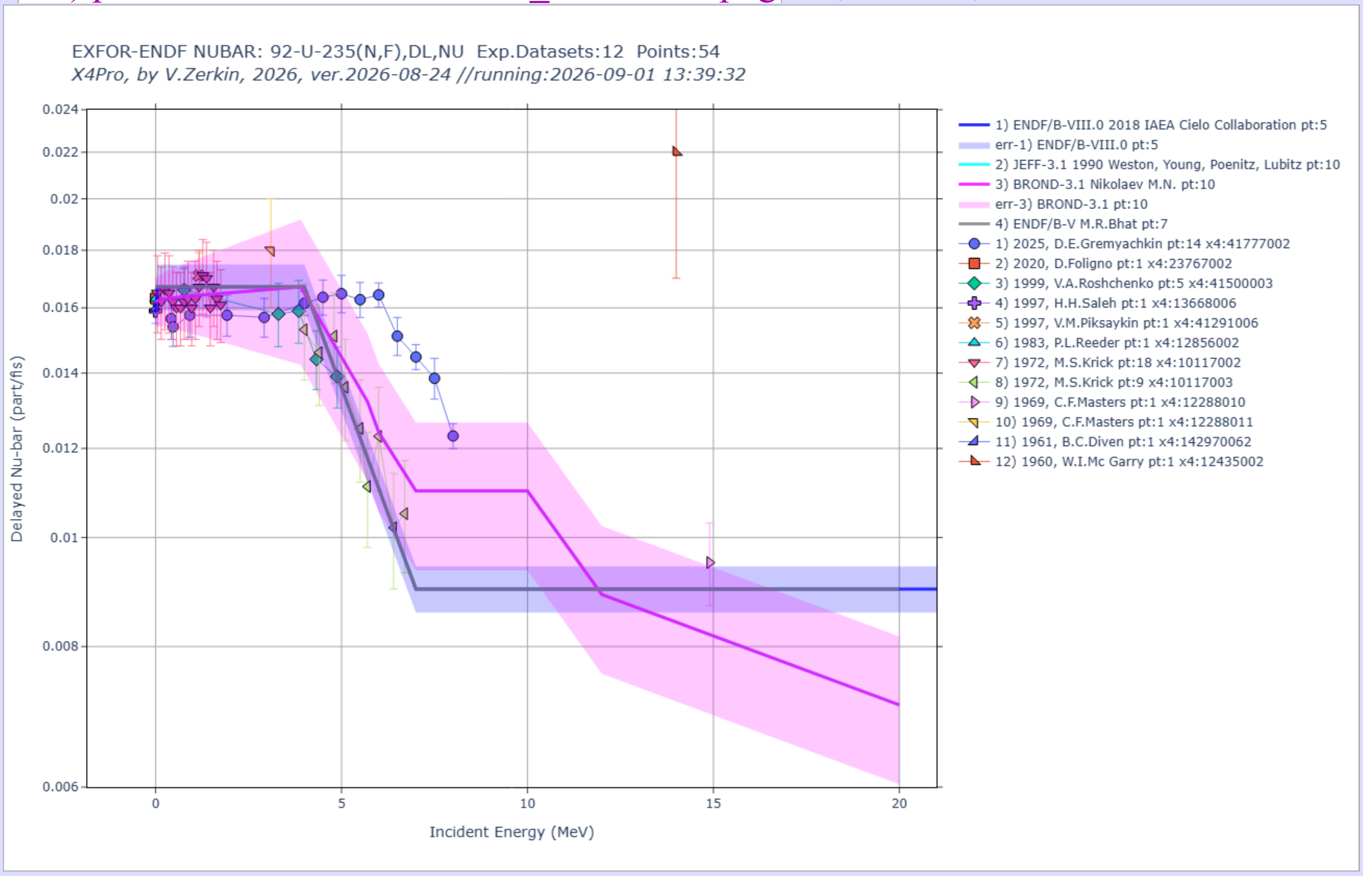
10) part2-9-nubar/out00/U235nf_-nu.html.png Total neutron yield (nu-bar)



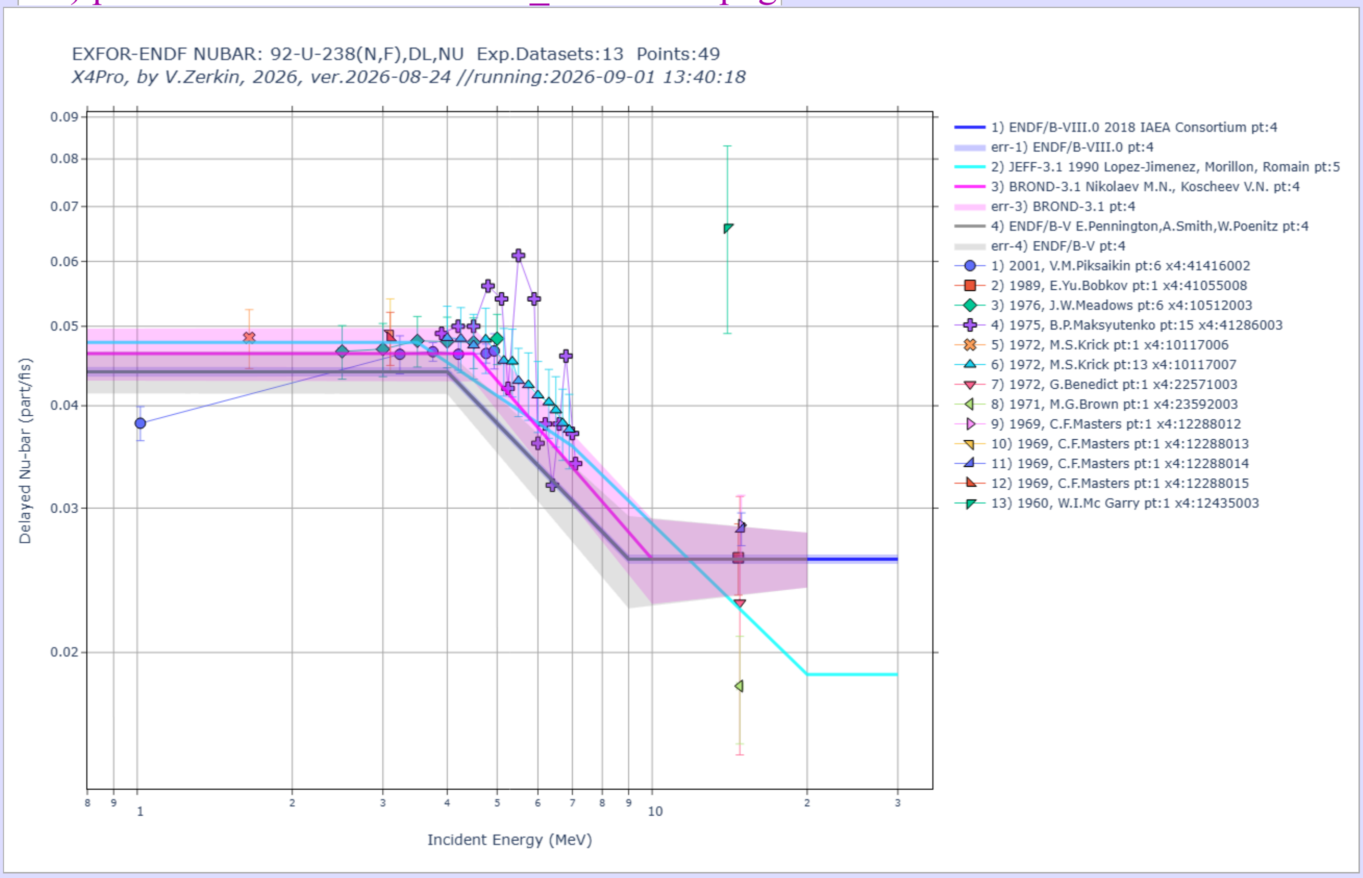
11) part2-9-nubar/out00/U235nf_pr-nu.html.png Prompt neutron yield (nu-bar)



12) part2-9-nubar/out00/U235nf_dl-nu.html.png Delayed neutron yield

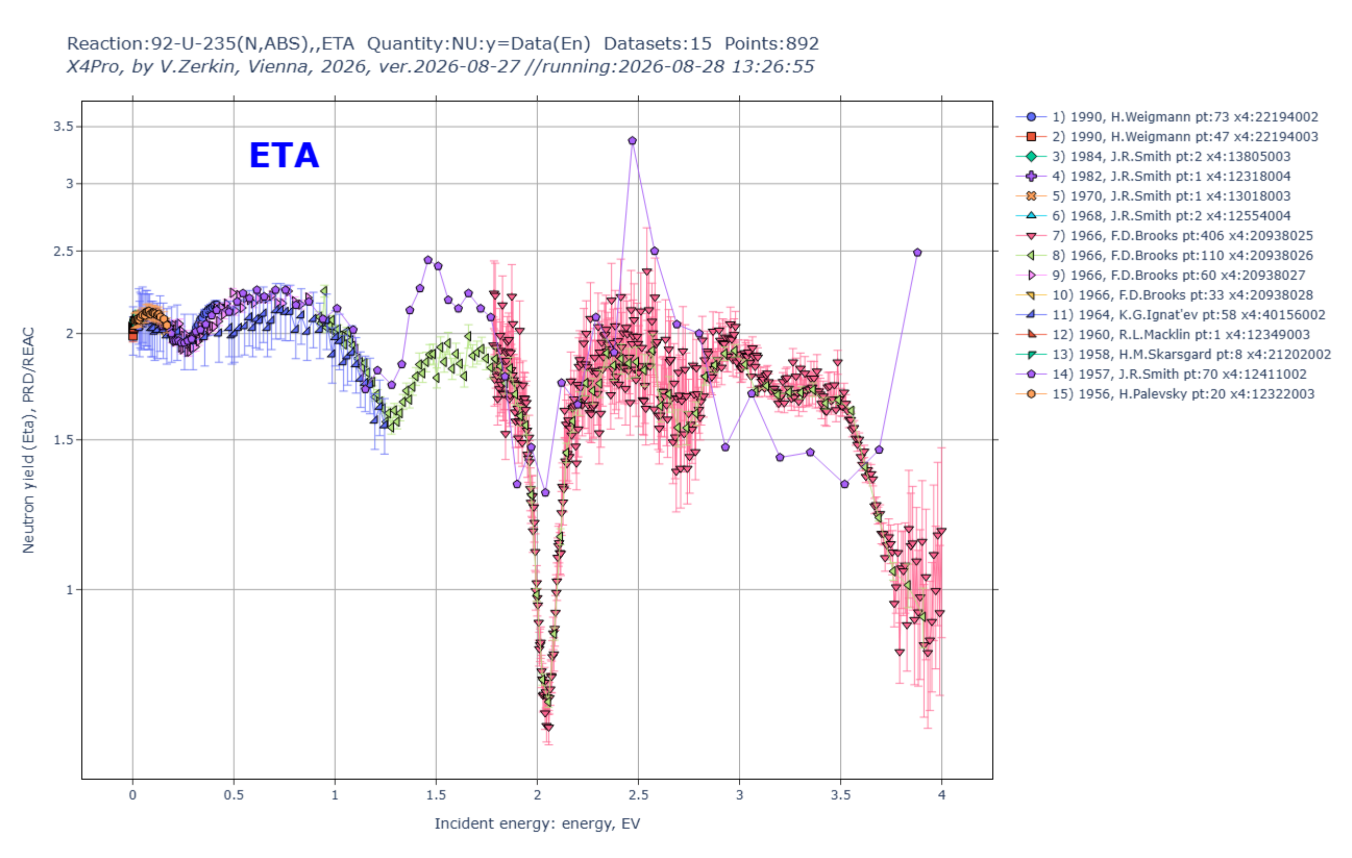


13) part2-9-nubar/out00/U238nf_dl-nu.html.png

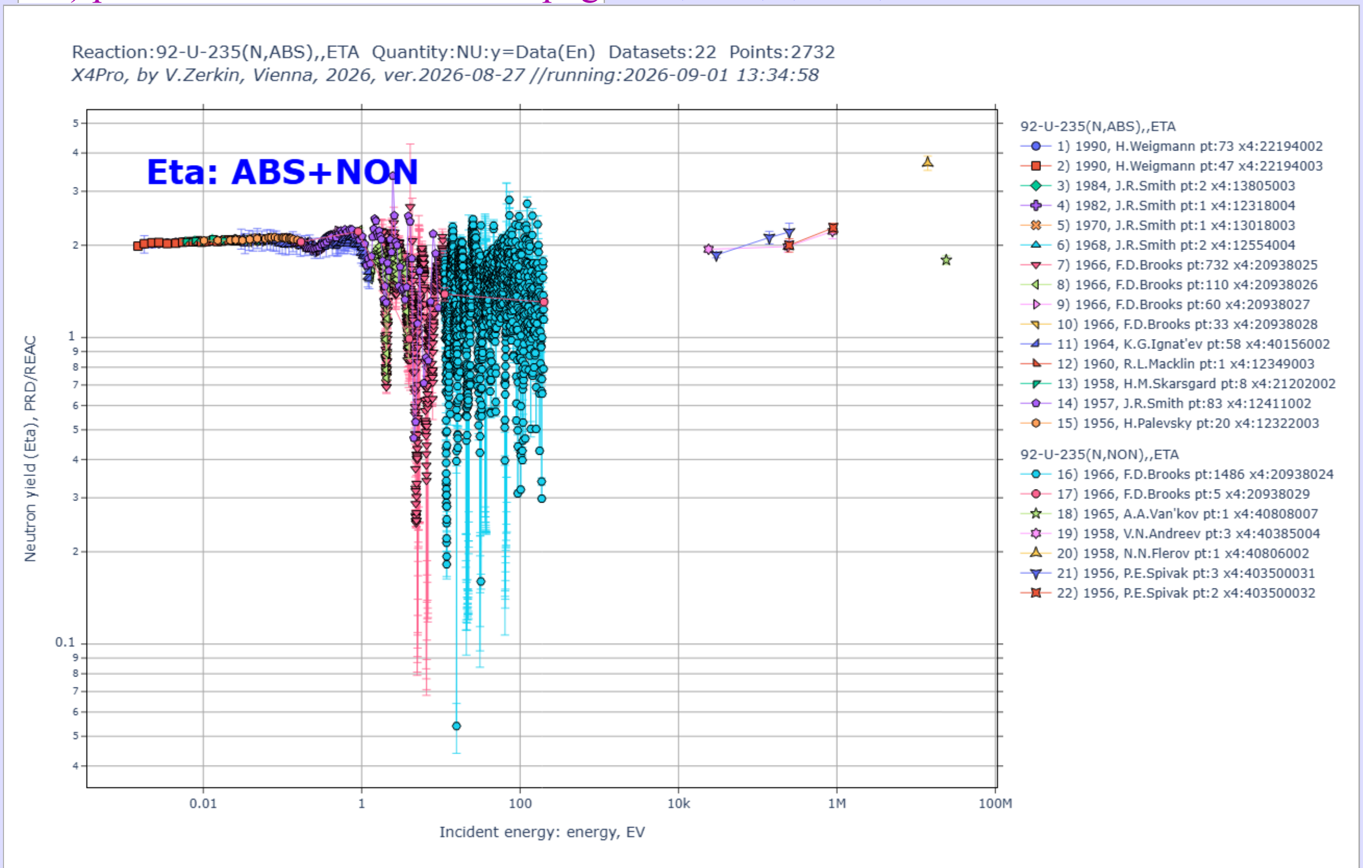


part1-8-reac/ *EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA*

14) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

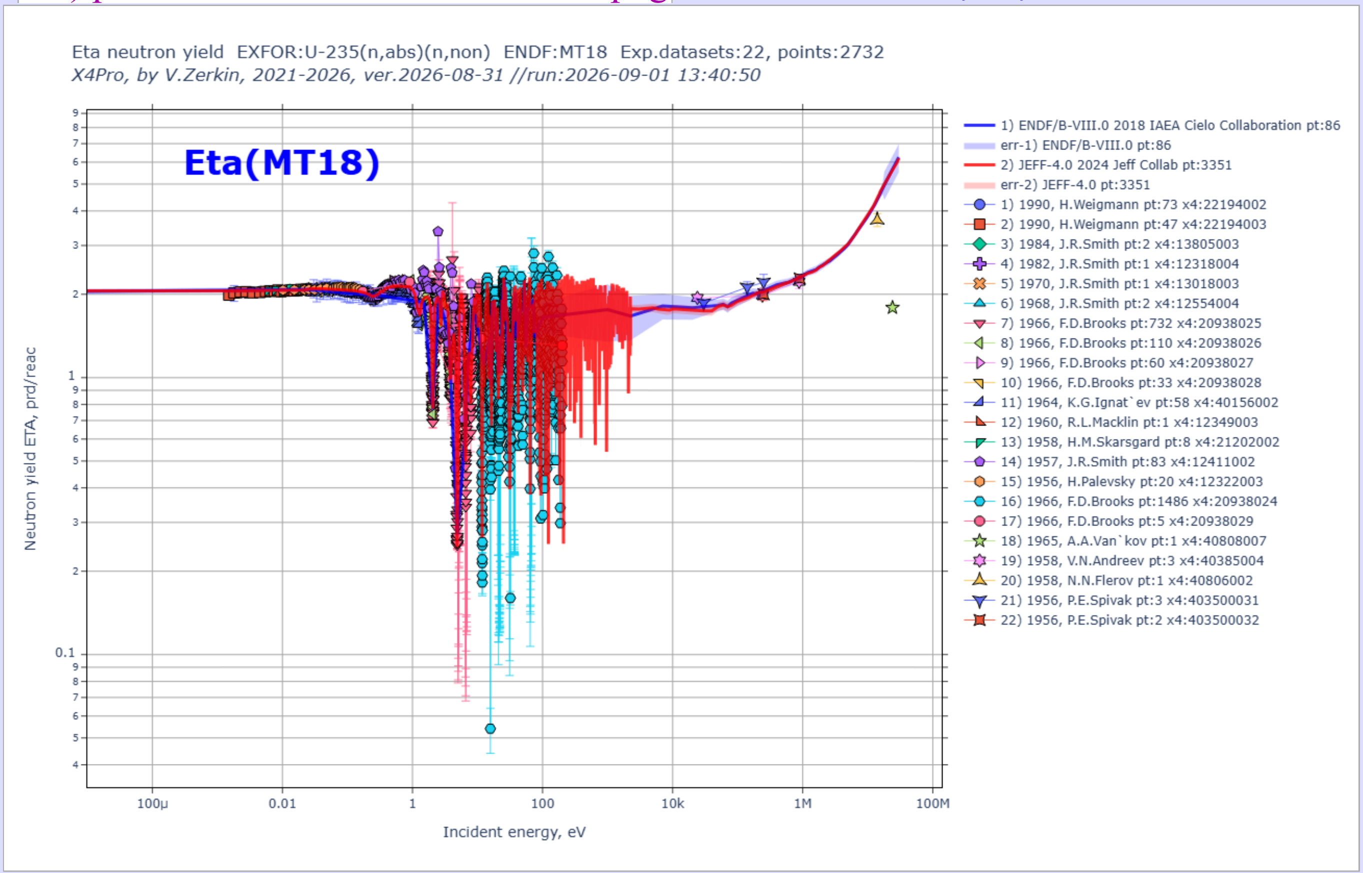


15) part1-8-reac/out00/eta2.html.png ETA: (N,ABS)+(N,NON)



part2-A-eta/ *EXFOR/ENDF Eta: neutron yield per nonelastic even*

16) part2-A-eta/out00/U235-eta.html.png ETA: EXFOR vs. ENDF η ± Δη



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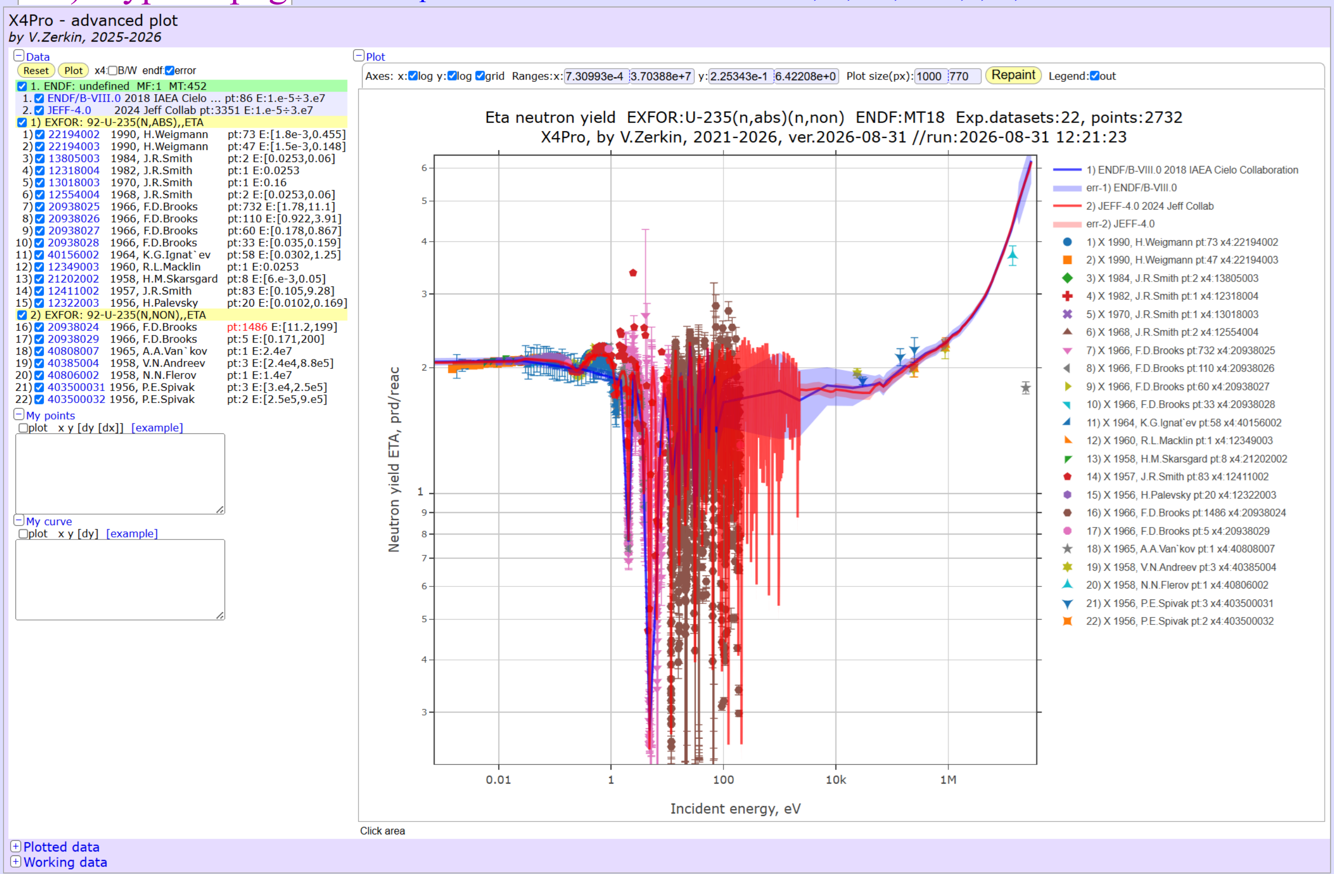
17) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT:452,18,102,MF:1,3,31,33



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